

AI-driven anomaly detection for public critical infrastructure resilience.

Public / Government sector × Critical infrastructure monitoring × Machine learning · OS073 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
46 is the world moving	76 can we play, can we win	MEDIUM 0 named in the evidence	€220m observed evidence · per year	LATER research and standards activi...	L0 13 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity is about deploying AI-driven anomaly detection to protect public critical infrastructure such as power grids, water treatment, and urban systems. It targets public sector operators who must ensure resilience against cyber threats and operational failures. The need is urgent because edge computing and AI have matured enough to make real-time, privacy-preserving detection feasible, and European research projects are actively validating these approaches.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Observed: tendered contract value, annualised	€220m €220m – €482m	€220m €220m – €482m	€13.2m €13.2m – €28.9m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

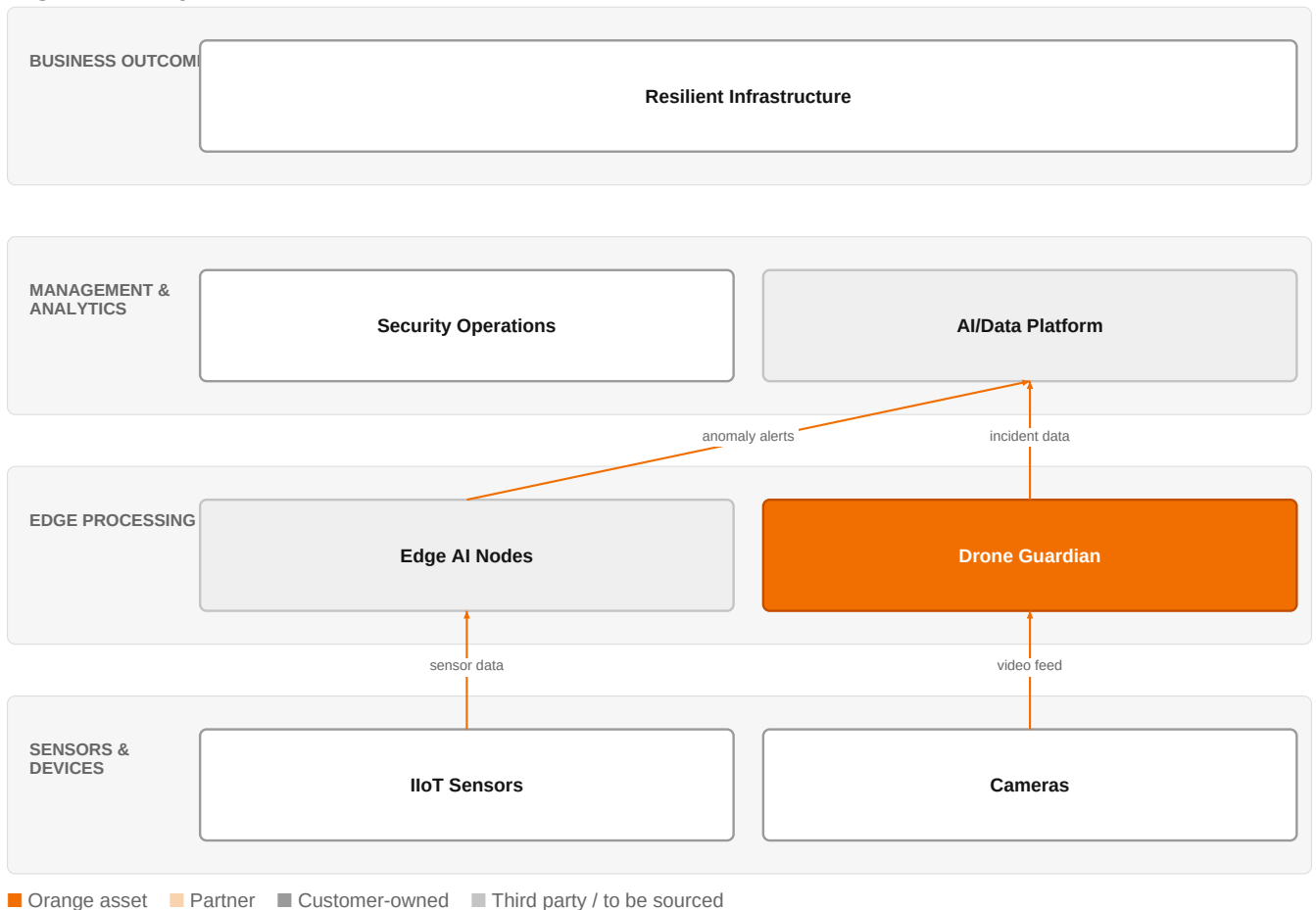
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€220m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	6.0 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 34 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

Edge AI Anomaly Detection for Critical Infrastructure



This solution detects anomalies at the edge, preserving data privacy while providing real-time alerts to security operations.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

The market is shifting because traditional signature-based and rule-based security cannot handle zero-day attacks or the scale of IoT traffic, as noted in recent research. Edge AI is emerging as a solution to latency and energy issues in cloud-based monitoring. Federated learning and privacy-preserving techniques are being developed to address data privacy concerns in industrial networks. European-funded projects like RESCUE are actively exploring resilient edge systems for critical infrastructure, signaling policy and research momentum.

Sources: SIG-8EC0D797B8D2 openalex.org 2025-10-27; SIG-F48074B55B20 openalex.org 2025-10-16; SIG-F633044AF87A openalex.org 2026-01-29; SIG-F292FB8C24B6 cordis.europa.eu 2025-09-01

Who buys, and why now

The buyer is typically a public sector agency or operator responsible for critical infrastructure, such as a national grid operator or a city administration. The pain is felt by operations and security teams who struggle to detect anomalies across complex sensor networks in real time, especially when data is sensitive and cannot leave the site. The trigger is often a near-miss incident, a regulatory audit, or a mandate to improve resilience against cyber threats.

Why it is hot now — every claim bound to a dated source

- The RESCUE project aims to strengthen the resilience, security, and adaptability of critical infrastructure systems through advanced edge computing, AI-driven anomaly detection, and robust communication networks. [SIG-F292FB8C24B6]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution combining its AI/Data/Cloud experts and Cybersecurity experts to design and deploy an edge-based anomaly detection system. The engagement would start with a security assessment and architecture

design, leveraging Orange's ISO 27001 and SecNumCloud certifications to ensure sovereignty. It would then deploy edge nodes running machine learning models, possibly using Orange Drone Guardian for physical perimeter monitoring, and integrate with partner clouds (AWS, Google Cloud) for scalability. The solution would be managed by Orange's Digital experts, with ongoing monitoring and tuning.

Why Orange can win

Orange can win because it combines sovereign, certified infrastructure (SecNumCloud, ISO 27001) with deep expertise in AI and cybersecurity, and has a proven reference with Grand Paris Sud. Unlike hyperscalers, Orange can offer a trusted, European sovereign solution. However, the asset list is thin on specific edge AI products, so Orange would need to leverage its research and partnership ecosystem to build a compelling offer.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Orange Drone Guardian	offer	L0	offer addresses the use case AND provides the technology	unconfirmed
AWS	partner	L2	partner provides the required technology	unconfirmed
Google Cloud	partner	L2	partner provides the required technology	unconfirmed
NICE	partner	L2	partner provides the required technology	unconfirmed
NVIDIA	partner	L2	partner provides the required technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SecNumCloud	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
CISPE	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
AI / Data / Cloud experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Cybersecurity experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Healthcare experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Grand Paris Sud	reference	SUP	published reference in this vertical, different use case	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 7.1; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitor	Category	Position here	Basis	Evidence
AWS	Hyperscaler	sells Machine learning; competes in Cloud — also an Orange partner	structural	—
Google Cloud	Hyperscaler	sells Machine learning; competes in Cloud — also an Orange partner	structural	—
Atos / Eviden	Global systems integrator	sells Machine learning; active in Public / Government sector; competes in Cloud, Cybersecurity	structural	—
NiCE	Customer-experience platform	sells Machine learning — also an Orange partner	structural	—
Databricks	Data and AI platform	sells Machine learning; competes in Cloud	structural	—
Dataiku	Data and AI platform	sells Machine learning; competes in Cloud	structural	—

Palantir	Data and AI platform	sells Machine learning; active in Public / Government sector; competes in Cloud	structural	—
Accenture	Global systems integrator	active in Public / Government sector; competes in Cloud	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What specific critical infrastructure do you operate, and what are the main anomaly types you need to detect (cyber, operational, safety)?
- 2 Do you currently have any anomaly detection in place, and what are its limitations in terms of latency, accuracy, or coverage?
- 3 Are you subject to any regulations or mandates that require enhanced resilience or sovereign data processing?
- 4 Where does your sensor data reside today, and are there constraints on moving it to the cloud?
- 5 What is the trigger for this initiative—an incident, an audit, or a strategic resilience goal?
- 6 Who would be the primary stakeholder for this project—operations, security, or IT?

Objections you will meet

They will say	You can answer
We already have a security operations center with traditional monitoring.	That's true, but traditional monitoring often misses zero-day attacks and complex anomalies. Our approach uses AI to detect patterns that rules can't catch, and we can complement your existing SOC.
We're concerned about data privacy and sovereignty.	We understand. That's why we offer SecNumCloud-certified infrastructure and can deploy edge AI that keeps data on-site, with federated learning to avoid centralizing sensitive data.
Why not just use AWS or Google Cloud for this?	They have strong platforms, but they may not meet your sovereignty requirements. Orange provides a European, certified alternative with the same scale, plus deep integration with your existing systems.

Risks and unknowns

The main risk is that the technology is still in research phase, with limited production deployments. Data privacy and regulatory constraints may slow adoption. The competitive landscape is crowded, and Orange's specific edge AI capabilities are not fully proven. Unknowns include the customer's existing infrastructure and willingness to adopt federated learning.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Assemble a bid angle on Critical infrastructure monitoring with Machine learning, combining Orange Drone Guardian (L0) and AWS (L2).
Sales	Open with Public / Government sector accounts on Critical infrastructure monitoring, leading with Orange Drone Guardian (L0) and the Grand Paris Sud (SUP) reference.
Strategist / Innovator	Commission a deep-dive on Critical infrastructure monitoring in Public / Government sector using Machine learning: size the opportunity bottom-up and decide whether to prototype next quarter.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-F633044AF87A	2026-01-29	openalex.org	1	SecuFL-IoT: an adaptive privacy-preserving federated learning framework for anomaly detection i...
SIG-0E742BED215C	2025-12-23	openalex.org	1	Improving Multivariate Time-Series Anomaly Detection in Industrial Sensor Networks Using Entrop...
SIG-E0AA4F59988C	2025-12-10	openalex.org	1	Anomaly detection for industrial applications: challenges, solutions, and future directions
SIG-3E11F4C71251	2025-11-13	openalex.org	1	Artificial Intelligence (AI) in Construction Safety: A Systematic Literature Review
SIG-8EC0D797B8D2	2025-10-27	openalex.org	1	AI-Powered Security for IoT Ecosystems: A Hybrid Deep Learning Approach to Anomaly Detection
SIG-F48074B55B20	2025-10-16	openalex.org	1	Implementation of edge AI for early fault detection in IoT networks: evaluation of performance ...
SIG-81F8AA75428C	2025-09-09	openalex.org	1	Time-Series-Based Anomaly Detection in Industrial Control Systems Using Generative Adversarial ...
SIG-F292FB8C24B6	2025-09-01	cordis.europa.eu	1	Resilient Edge Systems for Critical infrastructure and Urban Environments
SIG-5BF0E6E15A2F	2025-08-29	openalex.org	1	Decentralized AI on The Edge: Implementing Federated Learning for Predictive Maintenance in Ind...
SIG-B7834050E076	2025-08-26	openalex.org	1	A Case Study of Environmental Footprints for Generative AI Inference: Cloud versus Edge
SIG-565963A66A79	2026-06-01	arxiv.org	3	IstGPT: LLM-based Anomaly Detection for Spatial-Temporal Graph in Industrial Systems
SIG-2424954F4021	2026-05-24	arxiv.org	3	ASTRO: Adaptive Spatio-Temporal Reinforcement Optimization for GNN Powered Anomaly Detection in ...
SIG-17DA3B724138	2026-04-27	arxiv.org	3	System-aware contextual digital twin for ICS anomaly diagnosis

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:19+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:56:41+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.